

From NMF to D-SPIN

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1 Nonnegative Matrix Factorization for transcriptomics

This section is based on the *Learning the parts of objects by nonnegative matrix factorization* paper by D. D. Lee and H. S. Seung, expanded to cover genomics-related applications. The goal here so to extract and highlight the key points, while avoiding deep-diving into optimization algorithms, loss functions to be minimized and so on. The original paper is obviously much more detailed when it comes to that.

1.1 Image-related NMF

Let's take a collection of black-and-white images containing faces. Each picture contains a total on n pixels.

Each image can be flattened from 2 dimensions to 1 by collapsing it into a vector of n pixel intensities. Given m images, the result of the flattening is a matrix of positive values (pixel intensities cannot be negative)

$$Y \in \mathbb{R}_{\geq 0}^{n \times m}$$

The image database Y is regarded as an $n \times m$ matrix V , each column of which contains n nonnegative pixel values of one of the m facial images.

The goal of **Nonnegative Matrix Factorization (NMF)** is to decompose Y in a structure like

$$Y \approx HW$$

or, more explicitly,

$$Y_{i\mu} \approx \sum_{\alpha=1}^r W_{i\alpha} H_{\alpha\mu}$$

while imposing the **nonnegativity constraint**

$$H \in \mathbb{R}_{\geq 0}^{n \times r} \quad W \in \mathbb{R}_{\geq 0}^{r \times m}$$

The columns of W are called basis images. Each column of H is called an encoding, and is in one-to-one correspondence with a face in Y . An encoding consists of the coefficients by which a face is represented with a linear combination of basis images.

This nonnegativity constraint is what makes NMF different from other dimensionality reduction techniques like Principal Component Analysis. These constraints permit the combination of multiple basis images to represent a face, with the key point of only allowing additive combinations. In contrast to PCA, no subtractions can occur. For these reasons, the nonnegativity constraints are compatible with the intuitive notion of combining parts to form a whole, which is how NMF learns a part-based representation.

1.2 Bridging to genomics

Additive-only combinations are quite interesting when it comes to genomics, as a "negative" expression of a gene is something that cannot happen in nature and it is of difficult interpretation and understanding.

Therefore, we aim to build a correspondence which looks something like this

Image NMF	Transcriptomics
source image	sample / spatial location
pixels	genes
basis image	gene-expression program / factor
encoding of an image	factor activity in a sample/location

We can think of **gene program** as set or pattern of genes whose expression tends to vary together and that collectively reflects a biological process, cell state, or cellular function. In the specific context of NMF, the term usually refers more specifically to a learned factor characterized by a group of genes with high nonnegative weights, rather than to a predefined gene set.

1.2.1 Source images

In image decomposition, you start with an image such as

$$\boxed{\text{one image} = \text{one observation}}$$

Each row of your matrix is one face, i.e. a set of pixel intensities which is unique to a specific face

$$Y_i = [\text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_n].$$

In ordinary transcriptomics, we can make exactly the same construction. Suppose you have m cells and measure n genes. We go back exactly to

$$Y \in \mathbb{R}_{\geq 0}^{n \times m}.$$

but now each row is now one cell

$$Y_i = [\text{gene}_1, \text{gene}_2, \dots, \text{gene}_n].$$

The explicit analogy is

$$\boxed{\text{image} \longleftrightarrow \text{cell/sample}}$$

The image is a single observation whose features are pixel intensities.

The cell is a single observation whose features are gene-expression counts.

1.2.2 Pixels

images have pixel intensities

$$[\text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_n]$$

whereas a genomic sample has

$$[\text{gene}_1, \text{gene}_2, \dots, \text{gene}_n].$$

so the most direct analogy is

$$\boxed{\text{pixel} \longleftrightarrow \text{gene expression}}$$

In both cases, these are the **features** describing the observation.

There is a key difference to be noted. Pixels have a very obvious spatial relationship (e.g. pixel 100 is physically next to pixel 101), while genes generally don't have such a geometric relationship. Gene 100 is not necessarily "close" to gene 101 in any meaningful sense. When dealing with tissue samples where gene location is available, spatial information can be used. The process is an extension of NMF known as Nonnegative Spatial Factorization (NSF).

1.2.3 Basis images

Say that, for example, NMF discovers $r = 10$ basis images. For $j = 1, \dots, r$ we get

$$W_1 = [w_1^j, \dots, w_n^j]$$

Each W_j corresponds to a specific part of the faces: W_1 looks like an eye, W_2 looks like a nose, and so on. The key point is that a basis image is **not itself one of your original images**.

It is a learned pattern that can be used to reconstruct the original images. For one specific source image

$$\text{face} \approx 2.0(\text{eyes}) + 0.8(\text{nose}) + 1.3(\text{mouth})$$

The numbers 2.0, 0.8, 1.3 are the **encoding** for that particular face.

For transcriptomics, we replace "pixel intensity" with "gene expression". A basis image is therefore replaced by a **gene-expression pattern across genes**.

For example, suppose NMF discovers factor 1

$$W_1 = [0.9, 0.8, 0.7, 0.1, 0.0, \dots]$$

where the highly weighted genes are

$$[\text{GAD1}, \text{GAD2}, \text{SLC6A1}, \text{COL1A1}, \text{COL1A2}, \dots]$$

which are human genes that encode core proteins for GABAergic neurotransmission, the primary chemical messaging system that slows down and calms brain activity. Then factor 1 could be interpreted as an **inhibitory-neuron-associated gene program**.

Other factors might have high weights for another set of genes corresponding to different programs.

The key analogy here is

$$\boxed{\text{basis image} \longleftrightarrow \text{gene-expression program}}$$

1.2.4 Encoding

In image NMF, the encoding tells you **how much of each basis image is present in a particular image**.

Continuing with the previous example

$$\text{face}_i \approx 2.0(\text{eyes}) + 0.8(\text{nose}) + 1.3(\text{mouth})$$

The encoding is

$$H_i = (2.0, 0.8, 1.3)$$

In transcriptomics, the same thing happens. Suppose NMF finds three gene programs:

$$W_1 = \text{neuronal program}$$

$$W_2 = \text{immune program}$$

$$W_3 = \text{vascular program}$$

For cell i , NMF might give

$$H_i = (2.0, 0.1, 0.3)$$

which can be read this as

Cell/sample i has a strong contribution from the neuronal program, very little contribution from the immune program, and some contribution from the vascular program.

$$\boxed{\text{encoding} \longleftrightarrow \text{factor activity in a cell/sample}}$$

1.2.5 Final remarks

Image example:

$$\boxed{\text{image} \approx \text{encoding}_1 \times \text{basis image}_1 + \text{encoding}_2 \times \text{basis image}_2 + \dots}$$

Genomic experiment:

$$\boxed{\text{cell expression} \approx \text{activity}_1 \times \text{gene program}_1 + \text{activity}_2 \times \text{gene program}_2 + \dots}$$

Mathematically

$$Y_{i\mu} \approx \sum_{\alpha=1}^r W_{i\alpha} H_{\alpha\mu}$$

where

- $Y_{i\mu}$ = expression of gene μ in cell i ;
- $W_{i\alpha}$ = how strongly gene i belongs to gene program α ;
- $H_{\alpha\mu}$ = how active program α is in cell μ .

One subtle difference. NMF decomposes a face into

$$\text{face} = \text{eyes} \times c_{\text{eyes}} + \text{nose} \times c_{\text{nose}} + \text{mouth} \times c_{\text{mouth}} + \text{ears} \times c_{\text{ears}} + \dots$$

In transcriptomics,

$$\text{cell expression} = (\text{neuronal program}) \times c_{\text{neuronal}} + (\text{immune program}) \times c_{\text{immune}} + \dots$$

with different weights for each cell. The structure is the same, but we have to be careful with the analogy. **A gene program is** not literally a biological “part” in exactly the same sense as an eye is a physical part of a face. Rather, it is a **positive component of the observed expression profile**.

2 Nonnegative Spatial Factorization

This section is based on the *Nonnegative spatial factorization applied to spatial genomics* paper by F. W. Townes and B. E. Engelhardt, The goal here so to extract and highlight the key points, while avoiding deep-diving into optimization algorithms, loss functions to be minimized and so on. The original paper is obviously much more detailed when it comes to that.

2.1 Introducing spatial transcriptomics

Imagine two neighboring cells A and B , whose coordinates are very close. Biologically, it is often reasonable to expect their gene-expression patterns to be related. Ordinary NMF doesn't know that/makes no use of that information, i.e. *for ordinary probabilistic NMF, the factor for cell A and the factor for cell B are essentially treated as independent draws.*

The model knows

$$Y_i = \text{expression profile at location } i$$

but it doesn't explicitly know that

$$X_i \approx X_k$$

means we should expect

$$Y_i \text{ and } Y_k \text{ may be related}$$

Nonspatial models do not use the spatial coordinates x_i when defining the prior on the latent factors.

NMF asks

Can I explain these expression vectors using a small number of nonnegative gene-expression programs?

In spatial transcriptomics, we additionally know *where* each cell/location is. The question we pose ourselves is

Should the encoding of a spatial location depend on where that location is?

That is precisely the conceptual step from **NMF** to probabilistic NMF and to **Nonnegative Spatial Factorization (NSF)**.

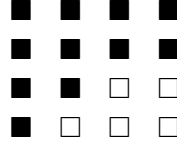
2.2 Using spatial information

Suppose that we have a tissue sample like

A	A	A	A
A	A	A	A
A	A	B	B
A	B	B	B

where A and B represent two different tissue environments.

We can expect a latent factor to vary smoothly over space, looking something like:



If two locations are close together, their factor values should tend to be similar. If two locations are far apart, they may be less similar.

2.2.1 Gaussian processes

A Gaussian Process (GP), in the most compact definition ever, is essentially a probability distribution over functions.

Imagine drawing a function $f(x)$ that, for our specific case, is a smooth factor-activity function. A gaussian process gives us a probabilistic model for what kinds of functions $f(x)$ are plausible.

GPs can encode the information

nearby spatial locations should tend to have similar values of the latent factor

through the so called **kernel** or **kernel function k** . Roughly, the kernel answers the question:

“How strongly should the factor values at locations x_i and x_k be correlated?”

If x_i and x_k are close $k(x_i, x_k)$ is large, if they are far apart $k(x_i, x_k)$ is smaller.

Instead of estimating each factor independently for every cell, we are effectively saying that *factor activity depends on spatial location*.

The paper on which this explanations is based on uses a Matérn kernel with smoothness parameter $3/2$. I’m still not sure deep diving into kernel functions is something I need to do.

2.2.2 NMF + GP = NSF

We can combine the two ideas.

NMF: gene expression $\approx$ factor activity $\times$ gene-factor association

GP: factor activity = a spatially correlated function

Nonnegative Spatial Factorization (NSF) simply combines these.

2.2.3 Mathematical basis

In transcriptomics, the analysis results are expressed in form of observed **counts**. For example:

$$Y_{i\mu} = 17$$

means 17 transcripts of gene μ were observed at location i .

Counts are naturally modeled using distributions such as the Poisson or negative binomial, so we can write

$$Y_{i\mu} \sim \text{Poisson}(\nu_i \lambda_{i\mu}).$$

with

- $y_{i\mu}$ = observed count
- $\lambda_{i\mu}$ = underlying expected expression level
- ν_i = size factor accounting for the fact that some locations simply have more total transcripts than others.

So far, nothing specifically spatial has happened.

The underlying expression level is what we are trying to model/describe.

$$\lambda_{i\mu} = \sum_{\alpha=1}^r W_{\mu\alpha} H_{\alpha\mu}$$

We now make the assumption that the number of transcripts is the same for all locations, so that we can just revert to the common form

$$Y_{i\mu} \approx \sum_{\alpha=1}^r W_{i\alpha} H_{\alpha\mu}$$

and then modify it to take into account spatial features. We will arrive at

$$Y_{i\mu} = \sum_{\alpha=1}^r W_{\mu\alpha} e^{f_{\alpha}(x_{\mu})}$$

$W_{\mu\alpha}$ is exactly the same as before, and it is still related to how strongly is gene μ associated with factor α .

The change is in the term that tells us how active is factor α at a certain spatial location.

There's one small thing we are missing. Gaussian processes have no positivity-related constraints by definition, and produce a real-valued quantity

$$f_{\alpha}(x_{\mu}) \in (-\infty, \infty).$$

but we want factor activity to be **positive**. Exponentiation is hereby used to transform

$$f_{\alpha}(x_{\mu}) \in (-\infty, +\infty) \quad \rightarrow \quad e^{f_{\alpha}(x_{\mu})} \in (0, +\infty).$$

so that the GP can model an unrestricted real-valued function $f_l(x)$ while the actual factor contribution $e^{f_l(x)}$ is always positive. The result is called an **exponentiated Gaussian process**.

The exponential guarantees $e^{f_{il}} > 0$, while $w_{jl} \geq 0$ is a "standard" constraint from NFM. Therefore every term in

$$Y_{i\mu} = \sum_{\alpha=1}^r W_{\mu\alpha} e^{f_{\alpha}(x_{\mu})}$$

is nonnegative, giving NSF its interpretable, parts-based structure while incorporating spatial information.

2.3 Practical example

Suppose there are $N = 12$ spatial locations and $r = 3$ factors. Let

$$Y_{i\mu}$$

denote the expression of gene i at spatial location μ .

A typical factor, let's say **factor 1**, of the NSF outcome looks something like

Spatial pattern	Genes
■ ■ ■ ■	Gene A : 0.9
■ ■ ■ □	Gene B : 0.8
■ ■ □ □	Gene C : 0.7
	Gene D : 0.1

and similarly for other factors.

2.3.1 NMF

$$Y_{i\mu} \approx \sum_{\alpha=1}^3 W_{i\alpha} H_{\alpha\mu}$$

For Factor 1, the example corresponds to

$$W_{\cdot 1} = \begin{pmatrix} 0.9 \\ 0.8 \\ 0.7 \\ 0.1 \end{pmatrix}$$

where the entries correspond to Genes A–D, and

$$H_{1\cdot} = (1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0)$$

If we reshape $H_{1\cdot}$ onto the 3×4 tissue grid, we obtain

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

The crucial point is that NMF treats the entries of $H_{1\cdot}$ as coefficients indexed by μ . It does not know that, for example, locations 1 and 2 are neighbors while locations 1 and 12 are far apart.

Consider the two vectors

$$H_{1\cdot}^{(A)} = (1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0)$$

and

$$H_{1\cdot}^{(B)} = (1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 0)$$

They contain the same number of nonzero entries and, if the corresponding columns of Y are permuted in the same way, they give exactly the same NMF reconstruction error. NMF has no

mechanism that prefers $H^{(A)}$ because its nonzero entries form one spatially coherent region.

More formally, for any permutation P of the spatial locations,

$$Y \approx WH$$

can be transformed into

$$YP \approx W(HP)$$

with exactly the same reconstruction error. Standard NMF is therefore invariant to permutations of the spatial locations. The spatial coordinates themselves do not enter the model.

So, to NMF,

$$(1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0)$$

and

$$(1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 0)$$

are just two possible coefficient vectors. Their spatial arrangement has no mathematical meaning to the factorization.

2.3.2 NSF

$$Y_{i\mu} = \sum_{\alpha=1}^3 W_{\mu\alpha} e^{f_{\alpha}(x_{\mu})}$$

For **factor 1**, the corresponding spatial contribution is

$$W_{\mu 1} e^{f_1(x_{\mu})}.$$

Here the factor is not simply an arbitrary vector of coefficients attached to the 12 locations. Its value is generated through the spatially indexed latent function

$$f_1(x_{\mu})$$

For the example, we could obtain a factor whose spatial expression is approximately

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

Thus the factor can be represented as

$$\begin{array}{cccc} \blacksquare & \blacksquare & \blacksquare & \blacksquare \\ \blacksquare & \blacksquare & \blacksquare & \square \\ \blacksquare & \blacksquare & \square & \square \end{array}$$

together with its gene-associated weights.

The distinction is that NSF has access to the coordinates x_{μ} , so the values

$$f_1(x_1), f_1(x_2), \dots, f_1(x_{12})$$

are values of a spatial latent function. Consequently, the model can distinguish between spatial configurations that NMF regards as equivalent.

For example,

$$f_1(x_\mu) \longrightarrow \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

represents a spatially coherent region, whereas

$$f_1(x_\mu) \longrightarrow \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

represents a spatially fragmented pattern.

Those two configurations are not equivalent under the NSF model because the spatial coordinates x_μ enter the latent function.

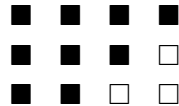
The cleanest statement is

NMF: $H_{1\cdot} = (h_{11}, \dots, h_{1N})$ is an arbitrary vector indexed by locations
 NSF: $e^{f_1(x_\mu)}$ is generated by a latent function of spatial location x_μ

Under NMF, permuting the spatial locations merely permutes the entries of H and does not change the mathematical status of the solution.

Under NSF, where those values occur in space is part of the model itself.

This is the sense in which NSF can distinguish a coherent pattern such as



from a spatially scrambled pattern with the same marginal factor values.

2.4 Hybrid Nonnegative Spatial Factorization

There is one last conceptual step we need to do. **Not every source of variation in gene expression is spatial**, nor two cells being neighbours implies similar gene expression between the two. This is the case, for example, of two neighboring cells whose gene expression is different because they are different cell types. Their difference will likely not be smoothly varying with position, as modelled by a GP.

Hybrid Nonnegative Spatial Factorization (NSFH) allows some factors to be spatial, others to be non spatial. Being T the number of spatial factors and $L - T$ the number of nonspatial factors. full NSF is recovered when all factors are spatial ($T = L$), while probabilistic NMF is recovered when none are spatial ($T = 0$).

3 Dimension-scalable Single-cell Perturbation Integration Network

This section is based on the "A guide to D-SPIN: constructing regulatory network models from single-cell RNA-seq perturbation data" paper by J. Jiang and M. Thomson, and on their prior works regarding D-SPIN.

D-SPIN (Dimension-scalable Single-cell Perturbation Integration Network) is a framework for learning probabilistic regulatory-network models from single-cell RNA-seq data collected across many perturbation conditions. Its central idea is to move from a representation of transcriptional states to a probabilistic model of how those states interact.

A natural starting point is non-negative matrix factorization (NMF), which decomposes the gene-expression matrix into latent components that can be interpreted as gene programs and their activities across cells. D-SPIN builds on this principle using orthogonal non-negative matrix factorization (oNMF) to identify positive gene programs that provide a compact representation of the expression landscape. The purpose of this factorization is therefore not yet to infer regulatory interactions, but to define a set of biologically interpretable variables on which such interactions can subsequently be modeled.

D-SPIN then takes a conceptual step beyond NMF by treating gene programs as nodes in a probabilistic graphical model. Rather than asking only which programs are active in a given cell, it asks whether the activity of one program is statistically associated with the activity of another across the population of cells. The distinction between program activity and program interaction is fundamental: the former describes the state of an individual cell, whereas the latter describes statistical dependencies between components across cells.

Perturbation conditions are incorporated into the same framework so that multiple experimental perturbations can be analyzed relative to a shared underlying interaction structure. In this formulation, perturbations act as condition-specific inputs that modify the tendency of particular programs to become active rather than requiring an entirely independent network to be inferred for every condition. This allows D-SPIN to integrate information from many perturbation experiments while maintaining a common representation of the regulatory system.

The resulting model consequently connects three levels of description: positive gene programs obtained from matrix factorization, their activity in individual cells, and the interactions that statistically couple their states. The transition from NMF to D-SPIN can therefore be explained as a transition from dimensionality reduction to probabilistic network inference: NMF provides the components with which transcriptional states are represented, whereas D-SPIN models how those components interact and respond to perturbations.

3.1 Basic architecture of D-SPIN

A schematic way of the principles behind D-SPIN is

genes $\longrightarrow$ gene programs $\longrightarrow$ program interaction network $\longrightarrow$ perturbation response

More explicitly, we can think of four objects:

W : gene-program composition
 H : program activities across cells
 J : interactions between programs
 B : effects of perturbations on the network

W and H make up the first layer (a factorization), while J and B constitute a probabilistic model for the program states.

3.2 Step 1: oNMF

The first layer finds positive gene programs that can constitute a basis for the observed expression patterns. NMF is used

$$Y \approx HW \quad Y \in \mathbb{R}_{\geq 0}^{n \times m} \quad H \in \mathbb{R}_{\geq 0}^{n \times r} \quad W \in \mathbb{R}_{\geq 0}^{r \times m}$$

with

n = number of genes
 m = number of cells/samples
 r = number of gene programs

D-SPIN imposes an additional orthogonality constraint, resulting in the so-called **orthogonal Nonnegative Matrix Factorization (oNMF)**. The additional restraint is applied to the activation/encoding matrix H . Here, the concept of matrix orthogonality, which is well-defined in the context of square matrices, is used in a "broader" definition for non-square matrices

$$H^T H = \mathbb{I}_r$$

Without getting into details, orthogonality encourages different programs to be separated rather than repeatedly explaining the same expression variation.

This is a modeling choice rather than a biological law. In real biology, the same gene can participate in multiple processes. Therefore, the non-overlap imposed by an orthogonal decomposition should be regarded as an interpretability constraint.

3.3 Step 2: From gene programs to program networks

Once the programs have been identified, each cell can be represented in a much smaller state space. Instead of n of gene-expression variables

$$G_1, G_2, \dots, G_n$$

we have a reduced number of programs r

$$P_1, P_2, \dots, P_r$$

For a particular cell i , let

$$\mathbf{s}^i = (s_1^i, \dots, s_k^i)$$

denote its program-level state (encoding).

At this point, an important conceptual change occurs. NMF asks:

How much of each program is present?

D-SPIN then asks:

Which combinations of program states are statistically favored?

The second question is a network question.

3.3.1 Networks

Suppose three programs have states s_a , s_b , and s_c . Imagine that, across a number of samples, states s_a and s_b are positively correlated

$$\mathbf{s}^1 = (s_a^1, s_b^1, s_c^1, \dots, s_k^1)$$

$$\mathbf{s}^2 = (s_a^2, s_b^2, s_c^2, \dots, s_k^2)$$

By correlation measure alone, we can not distinguish the three cases

$a \rightarrow b$ = program "a" activates program "b"

$b \rightarrow a$ = program "b" activates program "a"

$a \leftarrow c \rightarrow b$ = program "c" activates both programs "a" and "b"

Conditional-dependence model attempt to ask a more refined question

Does the relationship between A and B remain after accounting for the other variables?

3.4 Step 3. Programs interaction networks

D-SPIN uses a condition-dependent spin network, a particular type of Markov random field. The terminology comes from statistical physics.

Imagine that every gene or gene program is represented by a variable describing its state. The entire cell is then a configuration

$$\mathbf{s} = (s_1, \dots, s_K).$$

The model assigns a probability to every possible configuration. Borrowing from statistical mechanics, each s_i is the state of one microscopic degree of freedom, and the entire vector $\mathbf{s}$ is therefore a configuration or microstate.

3.4.1 Spin networks in statistical mechanics

In equilibrium statistical mechanics, the probability of observing a microscopic configuration is given by the Boltzmann distribution

$$p(\mathbf{s}) = \frac{1}{Z} e^{-\beta E(\mathbf{s})}$$

where $E(\mathbf{s})$ is the *energy of configuration*, $\mathbf{s}$, $\beta = 1/(k_B T)$ and Z is the so-called partition function

$$Z = \sum_{\mathbf{s}} e^{-\beta E(\mathbf{s})}$$

The crucial part is the definition of the energy of a single state. Without getting into details, the energy for a system of spins immersed in an external field is

$$-\beta E(\mathbf{s}) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j$$

where s_i can only assume the two opposite values ± 1 .

The term $h_i s_i$ describes the single-spin interaction with an external field (usually a magnetic field, when we talk about spins). Suppose $h_i > 0$. Then:

$$h_i(+1) = +h_i \qquad h_i(-1) = -h_i$$

Because the statistical-mechanical energy contains a minus sign, the $s_i = +1$ state has lower energy. We can think of h_i as a local preference for the spin in the presence of an external field.

The term $J_{ij} s_i s_j$ is a cross-interaction term between spins. The product $s_i s_j$ tells us whether two spins are aligned:

$$s_i s_j = \begin{cases} +1, & s_i = s_j \\ -1, & s_i \neq s_j \end{cases}$$

Therefore, the sign of J_{ij} determines what the system wants the two spins to do.

If

$$J_{ij} > 0$$

then the exponent is larger when $s_i = s_j$. Thus, the interaction favors alignment.

If

$$J_{ij} < 0$$

then the exponent is larger when $s_i \neq s_j$. Thus, the interaction favors anti-alignment.

$$\boxed{J_{ij} > 0 : \text{spins want to be parallel}}$$

$$\boxed{J_{ij} < 0 : \text{spins want to be antiparallel}}$$

3.4.2 Program networks in transcriptomics

When moving from spin systems to transcriptomics, we recycle the energy definition

$$-\beta E(\mathbf{s}) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j$$

but we have to be aware of what changes both in limits and interpretation.

The limits-related difference here is that spin now becomes something that describes the activity

or expression level of a gene programs in a given cell, and it not limited in its values like in the statistical mechanics framework

$$\begin{array}{ccc} s_i = \pm 1 & \longrightarrow & s_i \in \mathbb{R}_+ \\ \text{spin network} & & \text{programm network} \end{array}$$

The term $\boxed{h_i s_i}$ now relates to single-program activity tendency, while the cross-term $\boxed{J_{ij} s_i s_j}$ says that the state of program i affects the preference for the state of program j , describing both coordinated activity between two gene programs or their antagonistic relationship.

In the transcriptomic setting, the variable (s_i) is not a binary spin. Instead, it represents the activity of gene program (i) in a given cell, as inferred by oNMF in the specific case of D-SPIN. Consequently,

$$s_i \in \mathbb{R}_+$$

meaning that a program can have different degrees of activity, but its activity score is non-negative.

This difference is important when interpreting the interaction term. In the spin model, $(s_i s_j)$ can take both positive and negative values because each spin can be either $(+1)$ or (-1) . The sign of (J_{ij}) can therefore be interpreted directly in terms of whether two spins prefer to align or anti-align. This interpretation does not carry over directly to transcriptomic program activities, since

$$s_i \geq 0 \quad s_j \geq 0 \quad \implies \quad s_i s_j \geq 0 \quad \forall i, j$$

Here, (J_{ij}) instead determines how the simultaneous activity of two programs contributes to the statistical preference for a cellular state. For example, if

$$J_{ij} > 0,$$

then states in which both programs have high activity receive a larger contribution to the exponent

$$J_{ij} s_i s_j,$$

and the model therefore captures a tendency for the two programs to be active together. In this sense, a positive interaction is naturally associated with co-activity or coordination between the two programs.

Conversely, if

$$J_{ij} < 0,$$

then simultaneous high activity of the two programs contributes negatively to the exponent. The model therefore penalizes states in which both programs are strongly active at the same time, capturing a statistical tendency toward mutual exclusion or antagonistic activity.

Thus, for non-negative transcriptomic program activities, it is more appropriate to interpret

$$\boxed{J_{ij} > 0 : \quad \text{co-activation is statistically favored}}$$

$$\boxed{J_{ij} < 0 : \quad \text{simultaneous activation is statistically disfavored}}$$

rather than using the language of parallel and antiparallel spins.

The magnitude $(|J_{ij}|)$ describes the strength of this interaction within the model. Importantly, this interpretation concerns the joint statistical distribution of program activities. A positive

(J_{ij}) does not by itself imply that one program activates the other, just as a negative (J_{ij}) does not establish that one program represses the other. The inferred interaction describes a dependency in the observed cellular states; causal interpretation requires independent biological or experimental evidence.

3.5 Step 4. Introducing perturbations

3.5.1 Spin network

Let's go back to spins and statistical mechanics. We have our spin system and we now add an external perturbation. In statistical mechanics, a particularly simple perturbation is an additional external field acting on the spins. Suppose we apply a field b_i to spin i . The field contributes an additional term to the energy of the micro-state

$$-\beta E(\mathbf{s} | \mathbf{s}) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \sum_i b_i s_i$$

where we left the perturbation interact with different magnitude on each spin by letting b_i depend on i .

The perturbed distribution becomes

$$p(\mathbf{s} | \mathbf{b}) = \frac{1}{Z_{\mathbf{b}}} \exp \left[\sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \sum_i b_i s_i \right] = \frac{1}{Z_{\mathbf{b}}} \exp \left[\sum_i (h_i + b_i) s_i + \sum_{i < j} J_{ij} s_i s_j \right]$$

where we highlighted the fact that both the energy and the probability of state $\mathbf{s}$ are affected by the perturbation $\mathbf{b}$.

The perturbation does not change the interactions J_{ij} . Instead, it changes the external field experienced by the each spin

$$h_i \longrightarrow h_i + b_i$$

3.5.2 Transcriptomics

In transcriptomics,

$$\mathbf{s} = (s_1, \dots, s_K)$$

represents the state of the entire cell. The unperturbed model

$$p(\mathbf{s})$$

describes the distribution of cellular states under some baseline condition.

Now imagine experimentally perturbing the cell. For example, the perturbation could be a genetic knockout, activation of a transcription factor, inhibition of a signaling pathway, addition of a drug, or a change in the extracellular environment.

We therefore want to describe a conditional distribution

$$p(\mathbf{s} | q)$$

where q denotes the perturbation. The simplest way to represent perturbation q is to say that it produces an additional effective field on each gene or gene program i . We denote this field by

$$b_{iq} = \text{effect of perturbation } q \text{ on program } i$$

The perturbed distribution is consequently

$$p(\mathbf{s} \mid q) = \frac{1}{Z_q} \exp \left[\sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \sum_i b_{iq} s_i \right]$$

The sign of b_{kq} determines the direct effect of perturbation q on variable s_k :

$$b_{kq} \begin{cases} > 0 & \Rightarrow \text{favors larger } s_k \\ < 0 & \Rightarrow \text{favors smaller } s_k \end{cases}$$

Remarkably, even when $b_{iq} = 0$, the state s_i can still respond through the pairwise couplings J_{ij} .

This, the introduction of perturbations can be summarized as

perturbation $\rightarrow b_{iq}$ (direct) $\rightarrow J_{ij}$ (network-mediated) $\rightarrow$ cellular response

The parameter b_{iq} should be interpreted as the way in which perturbation q enters the model, rather than as a direct measurement of the biological effect of q on gene i . In this sense, b_{iq} represents an effective perturbation field: it describes the direct influence of the intervention on variable s_i . By contrast, the pairwise couplings J_{ij} describe how changes in one variable can propagate through the system. The resulting cellular response therefore emerges from the combination of the perturbation fields and the underlying interaction network.

This distinction is particularly useful when studying causality. By comparing the distribution of cellular states before and after a controlled perturbation, one can ask whether changing one component of the system produces a systematic change in another.

Example 1. Consider three gene programs,

$$s_1 = \text{interferon} \quad s_2 = \text{inflammation} \quad s_3 = \text{cell cycle}$$

Suppose the baseline model contains a positive coupling between the interferon and inflammation programs,

$$J_{12} > 0$$

This means that the model favors cellular configurations in which these two programs tend to be active together. Now consider a perturbation q that activates the interferon pathway. Suppose

$$b_{1q} > 0.$$

The perturbation directly pushes the interferon program toward larger values of s_1 . Because

$$J_{12} > 0,$$

the increase in s_1 can also make configurations with larger s_2 more probable. Thus, one could

observe

$$\Delta s_1 > 0$$

as a direct response and

$$\Delta s_2 > 0$$

as a network-mediated response. The model is therefore not simply saying that perturbation q changes gene program 1. Rather, it says that perturbation q changes the probability landscape of the entire cellular state through both its direct effects b_{iq} and the existing interactions J_{ij} .

Example 2. Consider two gene programs, s_A and s_B . In observational data, we may find that they are positively associated

$$\langle s_A s_B \rangle - \langle s_A \rangle \langle s_B \rangle > 0$$

However, this association alone does not establish that A regulates B . The two programs could, for example, be responding to a common upstream regulator.

Now suppose we experimentally perturb program A . This can be represented by a perturbation field b_{Aq} , while initially setting

$$b_{Bq} = 0$$

If the perturbation produces a systematic change in the distribution of s_B ,

$$p(s_B | q) \neq p(s_B)$$

then changing A has an effect on B . In the simplest interpretation, the causal chain is

$$A \xrightarrow{q} B$$

The pairwise model provides a mechanism for representing this response through the interaction network

$$A \xrightarrow{b_{Aq}} \text{network response} \xrightarrow{J_{AB}, J_{AC}, \dots} B$$

Thus, perturbations provide information that is not available from observational correlations alone: they actively change one part of the system and allow us to measure how the rest of the system responds. A controlled perturbation therefore helps distinguish *association* from *response to intervention*, which is the key step toward identifying causal relationships.

4 Introduction to the thesis use-case

4.1 Perturb-seq and single-cell genetic perturbation profiling

Understanding how genes control cellular phenotypes requires methods that can connect a specific genetic perturbation to its molecular consequences. Classical genetic screens have traditionally relied on relatively simple phenotypic readouts, such as cell viability, proliferation, or the expression of a selected marker. Although these approaches are highly scalable, they reduce the response of a cell to a small number of measurements and can therefore obscure the molecular mechanisms underlying a phenotype. Single-cell RNA sequencing (scRNA-seq), in contrast, provides a high-dimensional description of the transcriptional state of individual cells. Perturb-seq combines these two approaches by coupling pooled CRISPR-based genetic perturbations with single-cell transcriptomics, thereby enabling the molecular consequences of many perturbations to be measured simultaneously [?].

In a Perturb-seq experiment, cells are exposed to a library of single-guide RNAs (sgRNAs), with each guide targeting a particular gene or regulatory element. The perturbation identity is subsequently associated with the transcriptome of the same cell. Consequently, every sequenced cell can be regarded as an individual perturbation-response measurement: the sgRNA defines the experimental intervention, while the expression profile provides a high-dimensional phenotype. The original Perturb-seq implementation used an expressed guide barcode to connect the perturbation to the cell barcode and its corresponding transcriptome [?].

This experimental design provides an important conceptual advantage over bulk measurements. A perturbation does not necessarily produce a homogeneous response: cells carrying the same perturbation may occupy different transcriptional states because of stochasticity, cell-cycle differences, technical variation, or incomplete perturbation efficiency. These effects can be directly observed with single-cell measurements rather than being averaged away. At the same time, pooled screening substantially reduces the experimental cost compared with performing a separate RNA-seq experiment for every perturbation.

Perturb-seq is therefore particularly suited to the study of gene regulatory networks and genetic interactions. In a combinatorial experiment, cells can receive either one or multiple perturbations, making it possible to compare single-gene and multi-gene responses. The resulting data contain information not only about the effect of individual genes but also about how their effects combine. This is important because cellular phenotypes frequently arise from interactions between regulatory pathways rather than from the independent action of individual genes.

4.2 The Norman et al. Perturb-seq experiment

A particularly influential application of Perturb-seq was presented by Norman et al. in 2019 in the study Exploring genetic interaction manifolds constructed from rich single-cell phenotypes [?]. The authors investigated genetic interactions in human K562 cells using CRISPR activation (CRISPRa) and single-cell transcriptomic profiling. Unlike CRISPR knockout or CRISPR interference, CRISPRa is a gain-of-function approach: rather than reducing the expression of a target gene, it activates endogenous gene expression. This distinction is important when interpreting the resulting phenotypes, since the measured response corresponds to increased activity of the targeted gene or genes.

The experimental strategy was constructed in two stages. First, Norman et al. generated a large genetic interaction map using a growth-based CRISPRa screen. Pairs of genes were co-

activated and their effects on cellular fitness were compared with the effects expected from the corresponding single-gene perturbations. This allowed the authors to identify strong genetic interactions (GIs). However, a fitness measurement is intrinsically low-dimensional: two different molecular mechanisms can result in the same change in growth. Consequently, the fitness screen identified interactions without necessarily revealing why those interactions occurred.

Perturb-seq was subsequently used to provide a richer phenotype for a selected set of these interactions. Norman et al. profiled 132 gene pairs using combinatorial CRISPRa perturbations, together with the corresponding single-gene perturbations and controls. In total, the resulting experiment contained 287 perturbation conditions considered in the construction of the transcriptional genetic-interaction landscape [?].

The central conceptual contribution of the study was to regard each transcriptional phenotype as a point in a high-dimensional space. Instead of assigning a single scalar genetic-interaction score to a gene pair, the transcriptional response provides a vector containing the expression levels of many genes. The collection of these vectors was termed a genetic interaction manifold. Dimensionality-reduction techniques such as UMAP could then be used to visualize relationships between perturbations. Perturbations producing similar transcriptional states were positioned close to one another, allowing groups of genes and interactions to be associated with common biological processes [?].

This representation enabled several types of biological interpretation. Perturbations could be ordered according to their transcriptional similarity, allowing the authors to infer relationships between regulatory pathways. Furthermore, the rich phenotype could distinguish different mechanistic origins of genetic interactions that would appear similar in a conventional fitness-based screen. One notable result was the identification and characterization of a synergistic interaction involving CBL and CNN1, which was associated with erythroid differentiation [?].

The Norman et al. dataset therefore illustrates an important transition in functional genomics: genetic perturbations are not treated merely as labels for a binary phenotype, but as controlled interventions that probe the structure of a high-dimensional cellular state space.

4.3 Cutting corners

The present work uses a preprocessed version of the Norman et al. dataset available through Figshare [?]. This resource was published by Ethan Weinberger and is explicitly described as “Preprocessed CRISPRa Perturb-seq data from Norman et al., 2019”, prepared for the study introducing ContrastiveVI+ [?]. The original experimental data are available through GEO under accession GSE133344 [?].

The distinction between the original dataset and the Figshare resource is important. The Figshare object is not raw sequencing data and therefore does not require reconstruction of the complete experimental processing pipeline from sequencing reads. Instead, it provides a single-cell expression matrix together with cell-level metadata in a form suitable for computational analysis. In particular, the dataset is distributed as a processed single-cell data object and has been used in subsequent computational studies of perturbation responses [?].

The underlying Norman dataset contains cells subjected to zero, one, or two target-gene perturbations. Subsequent analyses of the GSE133344 data have reported approximately 110,000 cells and thousands of measured genes, with 236 distinct genetic perturbation conditions affecting 105 target genes in one processed representation of the dataset [?]. The exact number of cells and genes available to an analysis depends on the preprocessing and filtering steps applied, and

should therefore be reported explicitly for the particular Figshare file used in a thesis.

The processed dataset is particularly convenient for downstream single-cell analysis because the perturbation identity is represented in the cell metadata. This allows cells to be grouped according to their perturbation condition and compared with control cells. For example, cells carrying a non-targeting control can be used to characterize the baseline distribution of transcriptional states, whereas cells carrying a particular CRISPRa perturbation can be treated as samples from the corresponding perturbed distribution.

An important consideration is that the presence of an sgRNA does not necessarily imply that every cell has undergone an equally strong functional perturbation. Variable guide efficiency can generate cells that contain a perturbation label but exhibit little or none of the expected transcriptional response. This issue is particularly relevant when interpreting cell-level heterogeneity. Indeed, Weinberger et al. developed ContrastiveVI+ partly because pooled CRISPR screens contain noisy perturbation labels and because some guide-positive cells can escape the functional consequences of the perturbation [?]. Consequently, perturbation identity should be interpreted as an experimental assignment rather than as a perfect measurement of perturbation strength.

4.4 Application of D-SPIN to the Norman dataset

The preprocessed Norman et al. dataset is particularly suitable for D-SPIN because it contains many distinct perturbation conditions measured at single-cell resolution. A D-SPIN analysis can therefore use the perturbation identity in the cell metadata as the condition variable while using the expression matrix as the molecular phenotype.

A practical analysis consists of several conceptual stages. First, cells and genes should be quality-controlled and the expression matrix transformed into a representation appropriate for D-SPIN. The D-SPIN implementation expects log-normalized single-cell expression data and condition labels identifying the perturbation associated with each cell. Matched control cells are preferable because they allow perturbation responses to be estimated relative to the appropriate baseline [?].

Second, the expression space can either be represented directly at the gene level or reduced to gene programs. D-SPIN provides an unsupervised orthogonal non-negative matrix factorization (oNMF) procedure for identifying such programs, although predefined biologically motivated gene sets can also be used [?]. Program-level modeling is particularly attractive for a master-thesis analysis because it reduces dimensionality and often makes the resulting network easier to interpret biologically.

Third, D-SPIN jointly infers the interaction network and the perturbation-response vectors. The resulting network can then be used to determine which genes or programs interact, which perturbations act on particular parts of the network, and how different perturbations produce distinct cellular states. In the context of Norman et al., this provides a complementary perspective to the original genetic-interaction manifold: whereas Norman et al. primarily used transcriptomic similarity to organize perturbations, D-SPIN attempts to explain those observed responses through a shared network model.

This distinction also defines an important methodological advantage. Two perturbations can generate similar transcriptional profiles without acting on the same regulatory components, while two perturbations with apparently different profiles may affect different nodes of a common network. D-SPIN’s joint inference of network interactions and perturbation effects is designed to capture these relationships. The method has specifically been demonstrated on large Perturb-

seq datasets, including K562 perturbation data, where it was used to investigate regulatory organization and cell-fate responses [?].

Thus, the Norman et al. dataset and D-SPIN form a natural methodological combination. Perturb-seq provides the experimentally controlled, high-dimensional perturbation-response measurements; the preprocessed Figshare object provides these measurements in a computationally accessible form; and D-SPIN provides a probabilistic framework for integrating the different perturbation conditions into an interpretable network model. The result is a progression from perturbation to transcriptional phenotype and finally to network-level interpretation, which is the central rationale for using D-SPIN in the present study.

5 Dataset description

6 Analysis pipeline

7 Results